

ADITYA SINGH

AI/ML Engineer | LLMs · RAG · Agentic AI · NLP · Computer Vision

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PROFESSIONAL SUMMARY

Hands-on AI/ML Engineer with a B. Tech in Computer Science and practical experience building production-grade Agentic AI systems, LLM-powered pipelines, and multimodal ML solutions. Designed and deployed PolySensor — a real-world RAG-based content intelligence system processing 50+ document formats with an 80% reduction in manual review time. Strong foundation across the full ML lifecycle: data preprocessing, model training and evaluation, prompt engineering, retrieval-augmented generation (RAG), and agentic workflow design. Combines deep AI/ML engineering skills with a research-first mindset — always asking why a system behaves the way it does before optimising it.

TECHNICAL SKILLS

AI / LLM Stack	Large Language Models (LLMs), Prompt Engineering, Retrieval-Augmented Generation (RAG), Agentic AI, LangChain, LangGraph, Hugging Face Transformers, OpenAI API, Google Gemini API
ML & Deep Learning	PyTorch, TensorFlow, Scikit-learn, CNN, NLP, Computer Vision (YOLOv8 basics), Model Fine-tuning, Hyperparameter Tuning, Cross-validation, F1 / AUC / Precision / Recall
Data & Pipelines	Python, Pandas, NumPy, Matplotlib, Seaborn, ETL Pipeline Design, Data Preprocessing & Feature Engineering, SQL (SQLite, MySQL)
MLOps & Deployment	Flask (REST APIs), FastAPI (basic), Git, GitHub, Jupyter Notebook, Google Colab, Vercel (deployment), Docker (basic)
Multimodal & NLP	OCR (Tesseract), Speech Recognition, Document Intelligence, Text Classification, Sentiment Analysis, Named Entity Recognition (NER)
Research & Analysis	Wide & Deep Research Methodology, Hypothesis-driven Experimentation, ML Research Paper Reading & Implementation, Cross-domain Pattern Recognition

PROJECTS

PolySensor — Agentic AI Multimodal Content Intelligence System | *Personal Project* |

github.com/adityasinghcoding | polsensor.vercel.app

2023–2024

- Architected and deployed an end-to-end Agentic AI pipeline capable of ingesting and analysing content across 50+ file formats — documents, audio, video, images — using a multi-agent orchestration layer built on LangChain.
- Implemented an advanced RAG pipeline using Google Gemini as the base LLM with custom chunking strategies, vector store retrieval, and context-aware prompt engineering — achieving accurate cross-document reasoning on large corpora.
- Integrated OCR (Tesseract) and speech recognition modules for unstructured media extraction, feeding structured outputs into the RAG pipeline — enabling truly multimodal understanding in a single system.
- 80%** Optimised the full inference pipeline to reduce content review time by
- Deployed as a live web application on Vercel — demonstrating end-to-end ownership from architecture through production deployment.

Stack: Python, LangChain, Google Gemini API, RAG, OCR, Speech Recognition, Hugging Face, Vercel | Type: Agentic AI, Multimodal, Production Deployed

ACMA — AI Multimodal Content Moderation System | *Final Year Project* | github.com/adityasinghcoding

2024

- Designed and trained a multimodal content moderation system combining CNN (image/video) and NLP (text) models to detect toxic, inappropriate, and harmful content across media types.
- Achieved an F1-score of 0.91 through systematic hyperparameter tuning, class imbalance handling (weighted loss, oversampling), and ensemble evaluation across Precision, Recall, and AUC metrics.

- Built and deployed a real-time moderation REST API using Flask — enabling integration with external content pipelines and simulating a production content moderation service.
- Conducted thorough error analysis: reviewed false positives/negatives, identified model failure modes, and documented improvement hypotheses — practising the ML research iteration cycle.

Stack: Python, PyTorch, TensorFlow, CNN, NLP, Flask, REST API, HTML/CSS | Metric: F1 = 0.91

Enterprise AI Integration Architecture — Export Manufacturing | *Advisory Project — CL Gupta Exports Ltd.* 2024

- Designed a comprehensive AI integration architecture across 4 business functions for a mid-size export firm: HR anomaly detection (Isolation Forest), attrition prediction (XGBoost), customs document intelligence (LangChain + unstructured invoice parsing), demand forecasting (Facebook Prophet), and visual quality control (YOLOv8).
- Built an offline agentic document parser using pdfplumber and pyodbc — extracting structured records from ICEGATE customs shipping bill PDFs and inserting into Microsoft SQL Server without requiring an LLM, demonstrating pragmatic system design over technology-for-its-own-sake.
- Evaluated build-vs-integrate tradeoffs per function, selected architectures based on data availability, latency requirements, and implementation complexity — practising real-world ML system design thinking.

Stack: Python, pdfplumber, pyodbc, MS SQL Server, LangChain, XGBoost, Prophet, YOLOv8, Isolation Forest | Type: Agentic AI, MLOps, System Design

PROFESSIONAL EXPERIENCE

Machine Learning Intern | *CodSoft (Remote)*

Aug 2024

- Developed and evaluated three production-ready ML models across distinct problem types: Customer Churn Prediction (86% accuracy, identified 15% churn reduction opportunity), Movie Genre Classification (77% precision, multi-label NLP), SMS Spam Detection (97% accuracy, binary text classification).
- Preprocessed 50,000+ records using Pandas — handling nulls, encoding categoricals, engineering features, and optimising data loading to reduce training pipeline time by 20%.
- Applied systematic model selection: benchmarked multiple algorithms per task, evaluated using appropriate metrics (F1, Precision, Recall, AUC), and documented findings with reproducible Jupyter notebooks.

EDUCATION

Bachelor of Technology in Computer Science

2020 – 2024

Moradabad Institute of Technology (MIT), Moradabad, UP | GPA: 7.2 / 10 — First Division

Relevant Coursework: Artificial Intelligence, Machine Learning, Data Structures & Algorithms, Database Management Systems, Statistics

CERTIFICATIONS

- IBM Python 101 for Data Science — IBM (Mar 2024)
- Data Science & Machine Learning — ShapeMySkills (Dec 2023)
- Machine Learning — NPTEL / IIT Kharagpur (2023)

ADDITIONAL

- Continuous cross-domain research habit: independently studies ML research papers, AI system architectures, and emerging LLM techniques — translating findings into practical project implementations.
- Live deployed product: PolySensor is publicly accessible at polysensor.vercel.app — demonstrates production ownership, not just notebook-level work.
- 8× Gold Medalist in Speed Skating — structured discipline and goal-driven performance under competition.